12TH CHEMISTRY · CHAPTER 30 · SOLVED EXERCISE

Spectroscopy-2 NMR

Complete textbook exercise (MCQs, Short-Answer, Constructed Response, Descriptive) with Quick Check solutions

Q1. MULTIPLE CHOICE QUESTIONS

I. NMR relies on the magnetic properties of nuclei known as:

(a) Chemical shift (b) Nuclear density
(c) Nuclear spin (d) Signal intensity

Answer: (c) Nuclear spin — NMR is based on the nuclear spin of nuclei such as ^{1}H and ^{13}C.

II. Chemical shift (δ) in the ^{1}H NMR spectroscopy is measured relative to TMS and is expressed in:

(a) Tesla (T) (b) Hertz (Hz)
(c) Parts per million (ppm) (d) Degrees Celsius (ºC)

Answer: (c) Parts per million (ppm) — δ is measured in ppm from the TMS peak at 0.0 ppm.

III. Deuterated solvents ($CDCl_3$) are used in ^{1}H NMR to:

(a) Increase magnet strength (b) Split the signals
(c) Provide a reference signal (d) Avoid solvent signal interference

Answer: (d) Avoid solvent signal interference — Deuterium gives no peak in the ^{1}H spectrum, so the solvent is “silent”.

IV. The primary purpose of measuring the peak area in ^{1}H NMR is to find the:

(a) Chemical shift (b) Number of neighboring protons
(c) Total carbon atoms (d) Relative number of protons

Answer: (d) Relative number of protons — Peak area is proportional to the number of protons giving the peak.

V. Splitting is typically not observed for O-H or N-H protons because of:

(a) NMR-inactivity (b) Longer bond
(c) Deuterated solvent (d) Rapid chemical exchange

Answer: (d) Rapid chemical exchange — These protons are easily exchangeable, so they appear as singlets.

VI. In the n+1 splitting rule, ‘n’ represents the number of protons on:

(a) the entire molecule (b) the carbon being analyzed
(c) adjacent carbon atoms (d) the signal’s total area

Answer: (c) adjacent carbon atoms — n = protons on the adjacent (neighbouring) carbon atoms.

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VII. The chemical shift (δ) of a ^{13}C peak identifies the:

(a) Number of adjacent carbons
(b) Functional group environment (C=C, C=O, etc.)
(c) Molecule's symmetry
(d) Number of protons

Answer: (b) Functional group environment (C=C, C=O, etc.) — The position of the peak shows the type of carbon environment.

VIII. The number of peaks in a ^{13}C NMR spectrum equals the number of:

(a) Total carbon atoms
(b) Chemically non-equivalent carbons
(c) Adjacent hydrogen atoms
(d) ^{12}C atoms

Answer: (b) Chemically non-equivalent carbons — Each different carbon environment gives one peak.

IX. How many ^{13}C peaks does propane ($CH_3CH_2CH_3$) show?

(a) 1
(b) 2
(c) 3
(d) 4

Answer: (b) 2 — The two $-CH_3$ carbons are equivalent; the $-CH_2-$ carbon is different.

X. What is the splitting pattern for a proton with 2 neighbouring protons (n=2)?

(a) Singlet
(b) Doublet
(c) Triplet
(d) Quartet

Answer: (c) Triplet — n + 1 = 2 + 1 = 3 sub-peaks.

Q2. SHORT-ANSWER QUESTIONS

a) State two reasons why TMS is a standard for ^{1}H NMR.

(1) All 12 of its protons have the same chemical environment and produce a single sharp peak. (2) Its peak appears at the extreme right of the spectrum (low electronegativity of silicon), so it does not overlap with most organic proton signals. (It is also chemically inert and volatile, so it is easily removed.)

b) Why does deuterium (D) not appear in a ^{1}H NMR spectrum?

Deuterium requires a different frequency of radio waves (different energy) for spin flipping than ^{1}H. It does not absorb the radio frequency used for protons, so no peak appears for it.

c) A compound ($C_4H_{10}O$) has a ^{1}H NMR peak area ratio of 1:3:6. How many protons does each signal represent?

Total protons = 10 and the sum of the ratio = 1 + 3 + 6 = 10, so each unit of area is one proton. The signals represent **1 H, 3 H and 6 H**. (This fits 2-methoxypropane, $(CH_3)_2CH–O–CH_3$: CH, $O–CH_3$ and two equivalent CH_3 groups.)

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d) What is the purpose of the D_2O shake test in ^1H NMR?

It is used to detect whether a singlet in the spectrum is due to an O–H or N–H proton. After shaking with D_2O, the peak of the O–H or N–H proton disappears, which identifies it.

e) List the key features used to interpret a ^1H NMR spectrum.

1. Number of peaks: number of different proton environments.
2. Chemical shift (δ) of each peak: type of environment.
3. Area under each peak: relative number of protons in each environment.
4. Splitting pattern (n+1 rule): number of protons on the adjacent carbon atoms.

f) What does the number of peaks in a ^{13}C NMR spectrum represent?

The number of different carbon environments (chemically non-equivalent carbon atoms) in the molecule.

Q3. CONSTRUCTED RESPONSE QUESTIONS

a) A $-CH_2-$ group is lying between a $-CH_3$ and a $-CH<$ group. What is the splitting pattern for the $-CH_2-$ signal? Show your calculation.

$$n = 3 \text{ (from } -CH_3) + 1 \text{ (from } -CH<) = 4$$

$$\text{Number of sub-peaks} = n + 1 = 4 + 1 = 5$$

The $-CH_2-$ signal is split into five sub-peaks (a quintet/multiplet).

b) Predict the ^1H NMR spectrum of 1-butanoic acid.

Butanoic acid, $CH_3–CH_2–CH_2–COOH$, has four proton environments, so four peaks with area ratio 3 : 2 : 2 : 1.

Protons	δ (ppm)	Area	Splitting (n+1)
CH_3-	0.5 – 1.5	3	triplet (2 + 1)
$-CH_2-$ (middle)	0.5 – 1.5	2	6 sub-peaks (3 + 2 + 1; sextet)
$-CH_2-C=O$	2.0 – 2.6	2	triplet (2 + 1)
–COOH	9.0 – 13.0	1	singlet

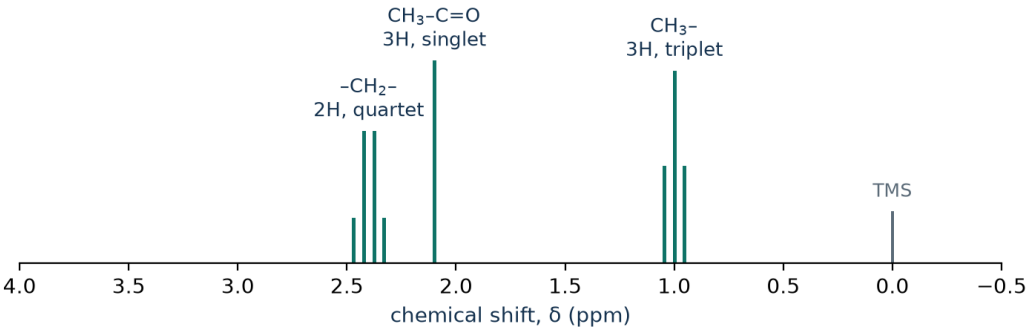
c) An organic compound has a molecular formula C_2H_5Cl. The ^1H NMR of the compound is given below. Deduce the structure of the compound by using its spectrum. Explain your working in detail. [Spectrum: a quartet at about 3.5 ppm and a triplet at about 1.5 ppm]

1. **Number of peaks:** two signals, so there are two proton environments.
2. **Chemical shift:** the signal near 3.5 ppm (3.0–4.5) is an alkyl group next to an electronegative atom (Cl); the signal near 1.5 ppm is an ordinary alkyl group.

3. **Splitting:** the quartet at 3.5 ppm has n = 4 − 1 = 3 neighbouring protons (it is next to a –CH_3); the triplet at 1.5 ppm has n = 3 − 1 = 2 neighbouring protons (it is next to a –CH_2–).
4. **Structure:** a –CH_3 joined to a –CH_2–Cl, i.e. **chloroethane, CH_3–CH_2–Cl** (peak areas 3 : 2). This is also the only possible structure for C_2H_5Cl.

d) An organic compound has a formula C_4H_8O. In laboratory, the compound gives 2,4-DNPH and iodoform tests positive but Fehling's test shows a negative result. The ^{1}H NMR of the compound is shown below. Deduce the structure of the compound and justify your answer. [Spectrum: 2H quartet at about 2.4 ppm, 3H singlet at about 2.1 ppm, 3H triplet at about 1.0 ppm]

1. **Chemical tests:** positive 2,4-DNPH shows a carbonyl group (aldehyde or ketone); negative Fehling's test shows it is not an aldehyde, so it is a ketone; positive iodoform test shows a methyl ketone, CH_3–CO–.
2. **Number of peaks:** three signals with areas 3 : 2 : 3 (total 8 H), so three proton environments.
3. **3H singlet at 2.1 ppm:** CH_3 next to C=O with no neighbouring protons (CH_3–CO–).
4. **2H quartet at 2.4 ppm:** –CH_2– next to C=O with three neighbouring protons (–CO–CH_2–CH_3).
5. **3H triplet at 1.0 ppm:** alkyl CH_3 with two neighbouring protons (–CH_2–CH_3).
6. **Structure:** CH_3–CO–CH_2–CH_3, **butan-2-one (butanone)**.



Sketch of the ^{1}H NMR spectrum of butan-2-one (own drawing)

DESCRIPTIVE QUESTIONS

Q4. A chemist suspects a sample contains an O-H group.

a) Describe the "D_2O shake" procedure.

b) What is the expected observation in the ^1H spectrum after the shake?

c) Write the equilibrium equation for the exchange and explain why it causes this observation.

a) Procedure

1. Record the ^1H NMR spectrum of the sample; it shows all the H atoms including the O–H proton.
2. Add a few drops of D_2O to the NMR sample tube and shake the solution well.
3. Record the spectrum again and compare the two spectra.

b) Observation

The singlet peak belonging to the O–H proton disappears in the second spectrum; all the other peaks remain.

c) Equation and explanation

Protons attached to highly electronegative atoms like O or N are easily exchanged. When D_2O is added, these hydrogens are quickly exchanged with deuterium atoms:

$$R–O–H + D_2O \rightleftharpoons R–O–D + HOD$$

Deuterium needs a different energy for spin flipping than ^1H, so it shows no peak in the ^1H NMR spectrum; it is "invisible".

Q.5 Predict the number of ^{13}C NMR peaks for the following two isomers:

a) Pentane

b) 2,2-dimethylpropane

Justify each answer by identifying the sets of equivalent carbons.

Isomer	Sets of equivalent carbons	Peaks
a) Pentane, $CH_3–CH_2–CH_2–CH_2–CH_3$	C-1 and C-5 (two CH_3); C-2 and C-4 (two CH_2); C-3 (middle CH_2)	3
b) 2,2-Dimethylpropane, $(CH_3)_4C$	four equivalent CH_3 carbons; the central quaternary carbon	2

QUICK CHECK QUESTIONS (IN-CHAPTER)

These questions appear in the "Quick Check" boxes inside the chapter, not in the end-of-chapter exercise.

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Quick Check 30.1

a) Explain why the 12 protons in TMS produce only a single peak.

In TMS, $Si(CH_3)_4$, the four methyl groups are attached to the same silicon atom in a symmetrical way, so all 12 protons have the same chemical environment (they are chemically equivalent). Equivalent protons have the same chemical shift, so only one sharp peak appears.

b) Identify how many chemical environments are there for protons in the following molecules. Also predict the expected chemical shift of the protons in each environment. i) 1,1,2,2-dichloroethane ii) propane iii) propan-2-ol iv) butan-2-one

"1,1,2,2-dichloroethane" is not a correct name (four locants need "tetrachloro"). It is solved as 1,1,2,2-tetrachloroethane, $CHCl_2$–$CHCl_2$; 1,2-dichloroethane, CH_2Cl–CH_2Cl, gives the same answer (one environment).

Molecule	Environments	Protons and expected δ (ppm), Table 30.1
i) $CHCl_2$–$CHCl_2$	1	H on carbon bearing Cl: 3.0 – 4.5 (or higher, as two Cl atoms are attached)
ii) CH_3–CH_2–CH_3	2	–CH_3 (6H): 0.5 – 1.5; –CH_2– (2H): 0.5 – 1.5
iii) CH_3–CH(OH)–CH_3	3	–CH_3 (6H): 0.5 – 1.5; >CH–O (1H): 3.0 – 4.5; O–H (1H): 0.5 – 6.0
iv) CH_3–CO–CH_2–CH_3	3	CH_3–C=O (3H): 2.0 – 2.6; –CH_2–C=O (2H): 2.0 – 2.6; –CH_3 (3H): 0.5 – 1.5

c) What proton environment does a signal at δ 7.5 ppm likely represent?

An aromatic proton (Ar–H, attached to an aromatic ring), whose range is 6.5 – 9.0 ppm.

Quick Check 30.2

a) What does the number of signals in a ^{1}H NMR spectrum represent?

The number of different chemical environments of protons present in the molecule.

b) Predict the number of peaks that will appear in the ^{1}H NMR spectra of the following molecules? (i) 1,1,2,2-dichloroethane (ii) butane (iii) propan-2-ol (iv) pentan-3-one (v) benzene

Molecule	Peaks	Reason
(i) $CHCl_2$–$CHCl_2$	1	both protons are equivalent
(ii) $CH_3CH_2CH_2CH_3$	2	two equivalent –CH_3 and two equivalent –CH_2– groups
(iii) $(CH_3)_2CHOH$	3	two equivalent –CH_3, the CH and the OH

Molecule	Peaks	Reason
(iv) $CH_3CH_2COCH_2CH_3$	2	two equivalent $-CH_3$ and two equivalent $-CH_2-$ groups
(v) C_6H_6	1	all six protons are equivalent

c) Predict the number of peaks and their chemical shifts in the ^{1}H NMR spectrum of bromoethane (CH_3-CH_2-Br).

Two peaks: $-CH_3$ protons at about 0.5 – 1.5 ppm and $-CH_2-$Br protons at about 3.0 – 4.5 ppm (next to the electronegative bromine).

d) How many signals will be observed in the ^{1}H NMR spectrum of 2-chlorobutane. Draw its structure and identify the hydrogen atoms which will show the highest value of chemical shift.

Structure: CH_3–CHCl–CH_2–CH_3. It has four proton environments (CH_3 next to CHCl, the CHCl proton, the CH_2 and the end CH_3), so **four signals**. The hydrogen on the carbon bearing chlorine (–CHCl–) shows the highest chemical shift (3.0 – 4.5 ppm).

e) What does the peak area in ^{1}H NMR represent?

The area under each peak is proportional to the number of protons giving that peak, so it shows the relative number of protons in each environment.

Quick Check 30.3

a) A signal in the ^{1}H NMR spectrum is a quartet. How many adjacent, equivalent protons ('n') are there?

$$n = \text{number of sub-peaks} - 1 = 4 - 1 = 3$$

b) Calculate the number of sub-peaks for each signal in the high-resolution spectra of the following compounds? i) butan-2-ol ii) 1-chloropropane iii) propanone iv) 2,2-dimethylpropane v) methanol

Compound	Signal	n	Sub-peaks (n+1)
i) CH_3–CH(OH)–CH_2–CH_3	CH_3 (on C-1)	1	2 (doublet)
	CH (on C-2)	3 + 2 = 5	6
	CH_2	1 + 3 = 4	5
	CH_3 (on C-4)	2	3 (triplet)
	OH	–	1 (singlet, not split)
ii) CH_3–CH_2–CH_2Cl	CH_3	2	3 (triplet)
	middle CH_2	3 + 2 = 5	6
	CH_2Cl	2	3 (triplet)

Compound	Signal	n	Sub-peaks (n+1)
iii) CH_3COCH_3	CH_3 (all six H equivalent)	0	1 (singlet)
iv) $(CH_3)_4C$	CH_3 (all twelve H equivalent)	0	1 (singlet)
v) CH_3OH	CH_3	0	1 (singlet)
	OH	–	1 (singlet)

Quick Check 30.4

a) Why do O-H or N-H peaks disappear after a D_2O shake?

The hydrogens of O–H and N–H are attached to highly electronegative atoms and are quickly exchanged with deuterium from D_2O (R–O–H becomes R–O–D). Deuterium needs a different energy for spin flipping, so it gives no peak in the ^1H NMR spectrum.

b) State the expected peak area and splitting patterns for ($-CH_2CH_2NH_2$).

Peak areas: 2 : 2 : 2 ($–CH_2–$: $–CH_2–N$: $–NH_2$). The $–CH_2–$ next to nitrogen is a triplet (two protons on the adjacent carbon; N–H protons cause no splitting). The other $–CH_2–$ is split into a triplet by the $CH_2–N$ protons (further splitting depends on the group attached to its other side). The $–NH_2$ protons give a singlet.

Quick Check 30.5

For the compound $(CH_3)_2CHOH$, predict the following: a) The number of peaks b) The type of proton and chemical shift c) The relative peak areas d) The splitting pattern

a) Three peaks (three proton environments).

Type of proton	b) δ (ppm)	c) Area	d) Splitting
two $–CH_3$ groups	0.5 – 1.5	6	doublet (1 + 1 = 2)
>CH–O	3.0 – 4.5	1	7 sub-peaks (6 + 1; septet)
O–H	0.5 – 6.0	1	singlet

Quick Check 30.6

a) How many signals do you expect to observe in the ^{13}C NMR spectrum of butane? Justify your answer.

Two signals. In $CH_3–CH_2–CH_2–CH_3$ the two end $–CH_3$ carbons are equivalent and the two $–CH_2–$ carbons are equivalent, so there are only two carbon environments.

b) Predict the ^{13}C NMR spectrum of butan-2-ol.

$CH_3–CH(OH)–CH_2–CH_3$ has four different carbon environments, so four peaks (unsplit lines): the C–OH carbon at 50 – 70 ppm and the three alkyl carbons (CH_3 on C-1, CH_2, CH_3 on C-4) at different positions within 0 – 50 ppm.

c) Predict the chemical shift of carbon atoms in propanoic acid.

Carbon in CH3–CH2–COOH	Environment	δ (ppm)
CH_3	alkyl	0 – 50
CH_2	next to carboxyl	30 – 65
COOH	carboxyl carbon	160 – 185

Quick Check 30.7

a) Draw the structures of butan-1-ol and butan-2-ol. Using the spectrum given in Figure 30.8, explain why this spectrum does not match with the structures of these two molecules.

Butan-1-ol: CH_3–CH_2–CH_2–CH_2–OH. Butan-2-ol: CH_3–CH(OH)–CH_2–CH_3. In each molecule all four carbon atoms are in different environments, so each would give **four** peaks. The spectrum in Figure 30.8 has only **three** peaks, so it cannot belong to either; it fits 2-methoxypropane, $(CH_3)_2$CH–O–CH_3.

b) Which peaks (mention chemical shift) would you expect in the spectra of butan-1-ol and butan-2-ol?

Butan-1-ol: one peak at 50 – 70 ppm (–CH_2–OH carbon) and three separate peaks within 0 – 50 ppm (two –CH_2– carbons and the –CH_3 carbon). **Butan-2-ol:** one peak at 50 – 70 ppm (>CH–OH carbon) and three separate peaks within 0 – 50 ppm (two –CH_3 carbons and the –CH_2– carbon).

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